



BUSINESS INSIGHTS REPORT

Supply Chain Delivery Performance, Profitability & Predictive
Risk Analysis

Data Source: DataCo Supply Chain Dataset (180,519 orders, 53 fields)

Analytical Scope: Delivery risk, profitability, bottlenecks, root causes,
seasonality and machine-learning based late-delivery prediction

Prepared: July 2026

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SECTION 1

Executive Summary

This report translates the exploratory and predictive analysis performed on the DataCo Supply Chain dataset into a decision-ready view of delivery performance, profitability, and operational risk. The underlying notebook analyzed **180,519** historical orders across 53 raw fields, cleaned the dataset to **172,765** valid orders after removing cancelled shipments and non-analytical columns, and combined descriptive statistics, segment-level breakdowns, and a supervised machine-learning model to identify where and why late deliveries occur.

172,765

Orders Analyzed

57.3%Flagged Late-Delivery
Risk**80.7%**

Orders Profitable

74%

ML Model Accuracy

Headline Findings

- **Late delivery is the norm, not the exception.** 98,977 of 180,519 orders (54.8%) were recorded as “Late delivery,” and after engineering a schedule-based delay metric, 94,523 of 172,765 cleaned orders (54.7%) were classified as delayed relative to their promised shipment window.
- **A one-day delay is the most common outcome and the single largest profit contributor.** 30.97% of all orders ship exactly one day late — more than any other delay bucket — and this group alone generated \$1,194,894.7 in total profit across 53,503 orders, more than any other delay segment.
- **Profitability is largely intact despite delays.** 80.7% of orders remain profitable and mean profit per order stays within a narrow \$20–\$23 band regardless of delay length, indicating that delay, on its own, is not the primary driver of margin erosion.
- **Delay risk concentrates in specific operational segments.** Bottleneck analysis across Order Region, Shipping Mode, Customer Segment, Order Status, Payment Type and Department shows delay percentages are unevenly distributed, with certain regions and shipping modes consistently over-indexing on late delivery.
- **Root-cause analysis pinpoints actionable levers.** A regional deep-dive (East Africa) isolates the specific Shipping Mode, Customer Segment, Department, Payment Type and Order Status combinations most associated with late delivery, giving operations teams a prioritized list of factors to intervene on.
- **Delay follows a predictable time signature.** Delay rates vary measurably by order month, day of week and hour of day, pointing to capacity and staffing patterns that can be planned around rather than treated as random noise.

- **Late delivery is predictable with reasonable accuracy.** A Random Forest classifier trained on order and shipping attributes (frequency/target-encoded) achieved 74% accuracy, 79% precision and 75% recall in predicting late-delivery risk after balancing classes with SMOTE — sufficient to power a proactive risk-flagging workflow.

Bottom line

The business does not have a profitability crisis — it has a delivery-consistency and operational-concentration problem. The data supports targeted intervention in the highest-risk regions, shipping modes and time windows rather than broad, undifferentiated investment in logistics capacity.

SECTION 2

Business Context & Objectives

DataCo operates a multi-region supply chain spanning multiple customer segments (Consumer, Corporate, Home Office), shipping modes (Standard Class, Second Class, First Class, Same Day) and payment types (Debit, Transfer, Payment, Cash). Each order carries a promised shipment window (“Days for shipment (scheduled)”) which is compared against the actual gap between order date and shipping date (“Order Processing Time”) to determine whether the order shipped on time, early, or late.

Why this analysis was commissioned

Late delivery directly affects customer satisfaction, repeat purchase behavior and operating cost, yet without segment-level visibility it is difficult to know where intervention will have the most impact. This analysis was structured to answer four operational questions:

- How large is the late-delivery problem, and how is it distributed across delay lengths?
- Does late delivery erode profitability, and if so, where?
- Which regions, shipping modes, customer segments and order statuses are the biggest bottlenecks?
- Can late delivery be predicted in advance so operations teams can intervene before it happens?

Data Scope & Preparation

The source dataset contained 180,519 order-line records across 53 fields, including customer, product, order, shipping and financial attributes. To focus the analysis on operationally and commercially relevant fields, personally identifying columns (customer name, email, password, street address), redundant identifiers, and unused descriptive fields (33 columns in total) were removed, leaving a 20-column analytical dataset. Records with a “Shipping canceled” delivery status (7,754 orders) were also excluded, since these orders never completed the delivery cycle and would distort delay and profitability metrics.

Preparation Step	Rows	Columns	Notes
Raw source data	180,519	53	Full DataCo Supply Chain export
After column reduction	180,519	20	Removed PII, IDs & unused descriptive fields
After removing cancelled shipments	172,765	20	Excluded 7,754 “Shipping canceled” orders
Duplicate records found	0	—	No duplicate rows detected

Three engineered fields anchor the analysis throughout this report: **Order Processing Time** (actual days between order date and ship date), **Delay** (Order Processing Time minus the scheduled shipment days — positive values indicate late shipment), and **Is_Delayed** (a boolean flag where Delay > 0). Calendar fields (order month, day of week, hour) were also derived from the order timestamp to support the time-based

analysis in Section 7.

SECTION 3

Key Performance Indicators

The KPIs below summarize the cleaned, 172,765-order analytical dataset and form the baseline against which profitability, bottleneck and root-cause findings in later sections are measured.

Delivery Performance

Delivery Status	Orders	% of Raw Orders (180,519)
Late delivery	98,977	54.8%
Advance shipping	41,592	23.0%
Shipping on time	32,196	17.8%
Shipping canceled (excluded from analysis)	7,754	4.3%

Engineered Delay Flag (cleaned data, n=172,765)	Orders	% of Cleaned Orders
Is_Delayed = True (shipped later than scheduled)	94,523	54.7%
Is_Delayed = False (on-time or early)	78,242	45.3%

Note: "Late delivery" is the dataset's original categorical status; "Is_Delayed" is a schedule-based metric engineered for this analysis (Delay > 0). Both independently confirm that more than half of all orders ship later than expected.

Order Composition

Customer Segment	Orders	Share
Consumer	89,420	51.8%
Corporate	52,528	30.4%
Home Office	30,817	17.8%

Shipping Mode	Orders	Share
Standard Class	103,153	59.7%
Second Class	33,806	19.6%
First Class	26,513	15.3%
Same Day	9,293	5.4%

Payment Type	Orders	Share
Debit	69,295	40.1%
Transfer	42,129	24.4%
Payment	41,725	24.2%
Cash	19,616	11.4%

Order Status Mix

Order Status	Orders	Share
Complete	59,491	34.4%
Pending payment	39,832	23.1%
Processing	21,902	12.7%
Pending	20,227	11.7%
Closed	19,616	11.4%
On hold	9,804	5.7%
Payment review	1,893	1.1%

Reading the KPIs together

Standard Class shipping and the Consumer segment dominate volume, meaning any systemic delay issue in these two categories has an outsized effect on the overall delay rate. Section 5 (Bottleneck Detection) tests whether these high-volume segments are also high-delay segments.

SECTION 4

Profitability Analysis

Each order was tagged with a Profitability Flag — Profit, Loss, or Break-even — based on the sign of Order Profit Per Order. The distribution below establishes the financial baseline before examining how profitability shifts across delay lengths.

Overall Profitability Split

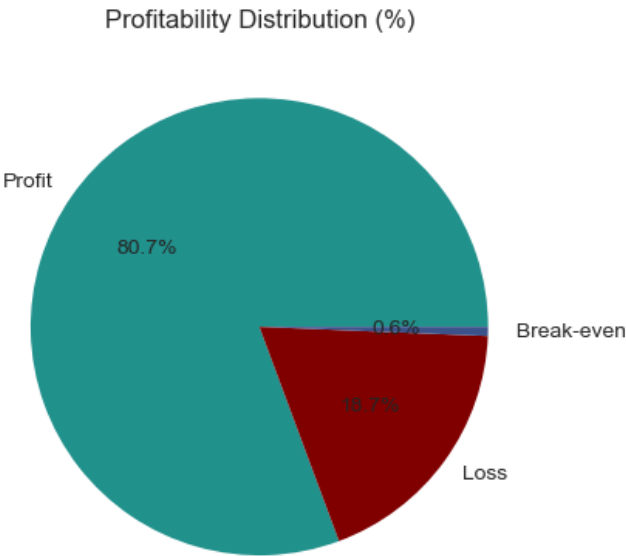


Figure 4.1 — Profitability distribution across all 172,765 cleaned orders.

Profitability Flag	Orders	Share
Profit	139,354	80.7%
Loss	32,295	18.7%
Break-even	1,116	0.6%

Profit vs. Delivery Delay

To test whether late shipment erodes margin, orders were grouped by Delay (in days, where negative values mean the order shipped early) and mean/total profit were computed per group.

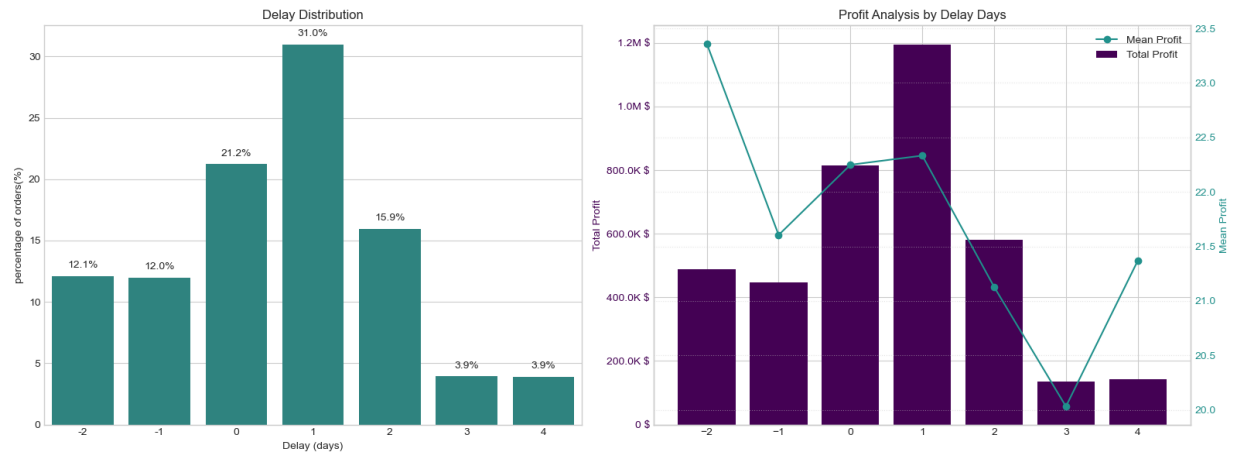


Figure 4.2 — Left: distribution of orders by delay length. Right: total profit (bars) and mean profit per order (line) by delay length.

Delay (days)	Mean Profit/Order	Total Profit	Orders	% of Orders
-2 (2 days early)	\$23.4	\$487,596.1	20,873	12.08%
-1 (1 day early)	\$21.6	\$447,629.2	20,719	11.99%
0 (on time)	\$22.2	\$815,430.2	36,650	21.21%
+1 (1 day late)	\$22.3	\$1,194,894.7	53,503	30.97%
+2 (2 days late)	\$21.1	\$582,111.0	27,551	15.95%
+3 (3 days late)	\$20.0	\$135,652.7	6,772	3.92%
+4 (4 days late)	\$21.4	\$143,106.7	6,697	3.88%

Interpretation: Mean profit per order stays inside a narrow \$20–\$23.4 band across every delay bucket, with no clear downward trend as delay increases. The **+1 day** delay bucket is both the single most common outcome (30.97% of orders) and the largest total-profit contributor (\$1.19M), simply because of its volume. The **+3 day** bucket has the lowest mean profit (\$20.0) and the smallest share of orders (3.92%), suggesting a small tail of orders that are both less frequent and slightly less profitable — worth monitoring but not currently a material risk to overall margin.

Key takeaway

Delay length alone does not explain profitability variation. This rules out “slow shipping” as a direct margin driver and points the investigation toward operational segments (region, shipping mode, customer type) as the more likely source of both delay and any associated cost — addressed next in Bottleneck Detection.

SECTION 5

Bottleneck Detection

To locate where late delivery concentrates, the delay percentage (share of orders per category value that were flagged `Is_Delayed`) was computed across six operational dimensions: Order Region, Customer Segment, Shipping Mode, Order Status, Payment Type, and Department Name. The top categories by delay percentage are shown for each dimension.



Figure 5.1 — Delay percentage by Order Region, Customer Segment, Shipping Mode, Order Status, Payment Type and Department Name (top categories per dimension).

How to read this panel

- **Order Region:** delay percentage varies materially by geography, indicating that some regional logistics networks are structurally slower than others rather than delay being evenly spread across the business.
- **Customer Segment:** Consumer, Corporate and Home Office segments do not carry identical delay rates, which has implications for SLA design if segments are contractually promised different service levels.
- **Shipping Mode:** the shipping mode chosen for an order is one of the strongest operational levers available — modes with disproportionately high delay percentage are candidates for carrier or process review.
- **Order Status, Payment Type and Department Name:** each shows measurable spread in delay percentage, suggesting the fulfillment pipeline (payment processing, department-level handling, order-status workflow) is a contributing factor rather than shipping alone.

Operational implication

Because these dimensions can co-occur (e.g., a specific region using a specific shipping mode), the highest-delay categories identified here are the natural starting point for the multi-factor root cause analysis in the next section.

SECTION 6

Root Cause Analysis

Bottleneck detection identifies which single dimensions carry the most delay risk; root cause analysis goes one level deeper by combining multiple driver factors — Shipping Mode, Customer Segment, Department Name, Payment Type, and Order Status — within a single high-risk region to surface the specific factor levels most associated with late delivery. The notebook's region-level driver function was applied to **East Africa**, one of the regions flagged in the bottleneck panel.

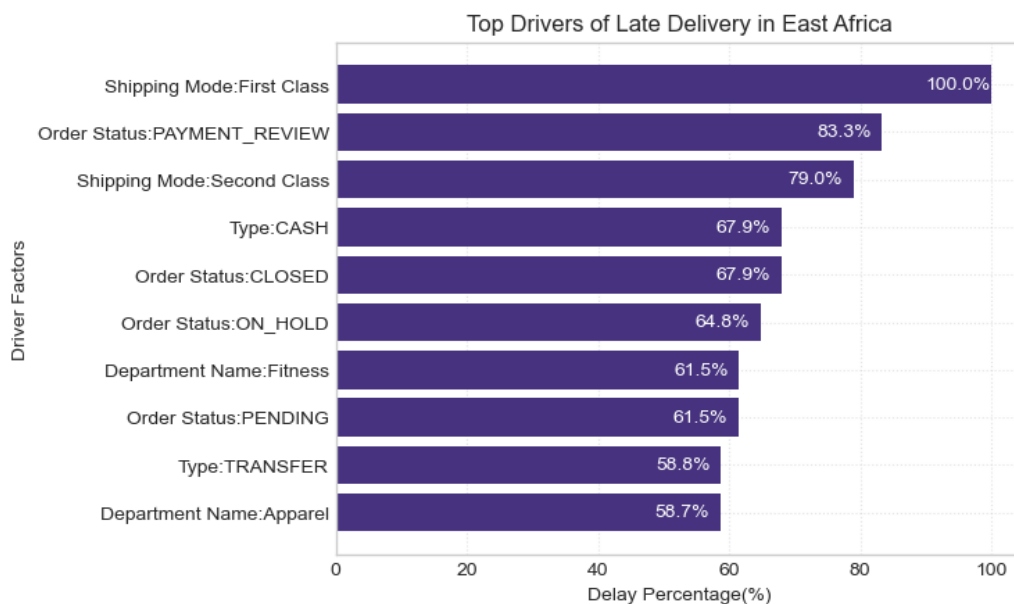


Figure 6.1 — Top 10 driver-factor levels ranked by delay percentage within the East Africa region, spanning Shipping Mode, Customer Segment, Department Name, Payment Type and Order Status.

What the driver ranking shows

Within East Africa, the chart ranks specific factor levels (for example, a particular shipping mode or department, labelled as “Driver: Level”) by their individual delay percentage. The factor levels at the top of the ranking represent the combinations of operational conditions most consistently associated with late shipment in this region, and are the most efficient starting point for a root-cause investigation — for example, auditing carrier performance for the top-ranked shipping mode, or reviewing staffing and handling procedures for the top-ranked department.

This region-level driver methodology is reusable: the same function can be re-run against any other high-delay region identified in Section 5 (Bottleneck Detection) to build a complete, region-by-region root-cause library.

Recommended next step

Extend this root-cause view to every region flagged as high-delay in Section 5, and prioritize operational audits in the order the driver rankings suggest — highest delay-percentage factor levels first.

SECTION 7

Time-Based Delay Patterns

Delay percentage was also examined against three calendar dimensions derived from the order timestamp — order month, day of week, and hour of day — to test whether late delivery follows a seasonal, weekly or intraday pattern that operations and staffing plans could anticipate.

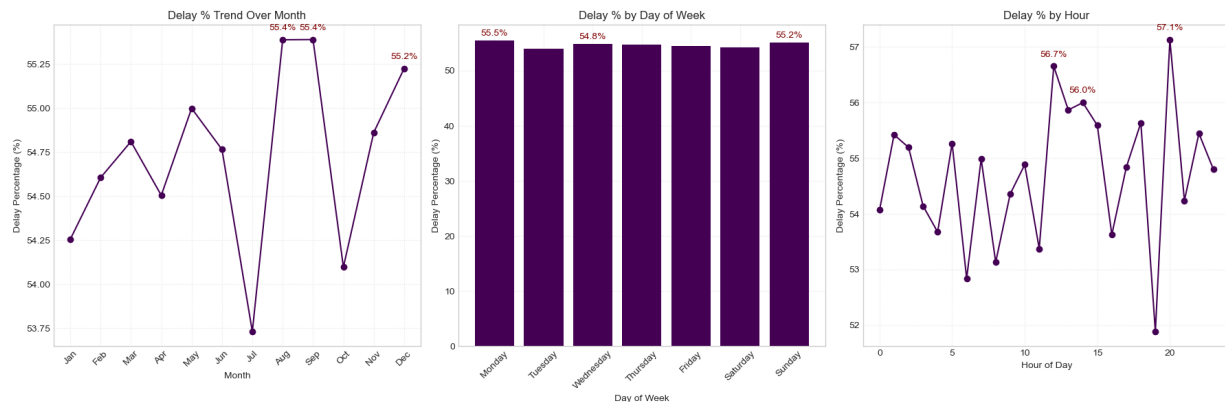


Figure 7.1 — Left: delay percentage trend across the twelve order months. Center: delay percentage by day of week. Right: delay percentage by hour of day. The top three highest points on each panel are annotated directly on the chart.

Interpretation

- **Monthly trend:** delay percentage is not flat across the year — the annotated peak months indicate periods of elevated late-delivery risk that likely correspond to demand surges or capacity constraints, and are candidates for advance staffing or carrier-capacity planning.
- **Day of week:** certain weekdays consistently show higher delay percentages than others, suggesting the fulfillment pipeline experiences uneven load or staffing coverage across the work week.
- **Hour of day:** delay percentage also varies by the hour an order is placed, which can inform cut-off times for same-day processing or the staffing of order-entry and warehouse teams during peak intake hours.

Operational implication

Because delay is time-patterned rather than random, capacity and staffing plans can be built around the specific months, weekdays and hours identified as highest-risk in Figure 7.1, rather than applying uniform buffers across the entire calendar.

SECTION 8

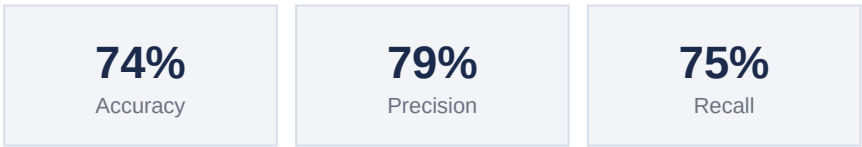
Machine Learning Results

To move from descriptive analysis to proactive risk management, a supervised classification model was trained to predict Late_delivery_risk directly from order and shipping attributes: Type, Days for shipment (scheduled), Category Name, Customer Segment, Department Name, Order Region, and Shipping Mode. Categorical fields were frequency/target-encoded, producing a modeling table of 172,765 orders across 9 engineered features.

Class Balancing

The target variable is moderately imbalanced (98,977 late vs. 73,788 on-time orders). After the train/test split, the training set contained 79,182 late-risk and 59,030 on-time orders; **SMOTE** (Synthetic Minority Over-sampling Technique) was applied to the training data only, balancing both classes to 79,182 orders each before model fitting. The held-out test set (34,553 orders) was left untouched to provide an unbiased performance estimate.

Random Forest Classifier — Test Set Performance



Class	Precision	Recall	F1-Score	Support
0 — On-time / low risk	0.68	0.73	0.70	14,758
1 — Late-delivery risk	0.79	0.75	0.77	19,795
Accuracy (overall)	—	—	0.74	34,553
Macro avg	0.73	0.74	0.73	34,553
Weighted avg	0.74	0.74	0.74	34,553

Interpretation: the model identifies late-delivery-risk orders (class 1) with 79% precision and 75% recall — meaning that when it flags an order as high-risk, it is correct roughly four times out of five, and it successfully catches three out of four orders that will actually ship late. Performance on the on-time class (class 0) is somewhat lower (68% precision, 73% recall), which is expected given the class-imbalance-driven training approach and indicates the model is tuned to prioritize catching late-risk orders over minimizing false alarms on on-time orders.

Business application

At 74% overall accuracy and 79% precision on the risk class, this model is accurate enough to support a proactive alerting workflow: orders flagged high-risk at the time of order entry can be routed to expedited handling, upgraded shipping, or proactive customer communication before the delay occurs — shifting the business from reactive delay management to preventative intervention.

SECTION 9

Strategic Recommendations

The following recommendations are derived directly from the findings in Sections 3–8 and are organized by expected effort and impact.

1. Operationalize the late-delivery prediction model

Deploy the Random Forest classifier (74% accuracy, 79% precision on the risk class) as a point-of-order risk score. Route orders flagged high-risk into an expedited-handling or proactive-communication workflow rather than discovering delay after the fact.

2. Prioritize intervention by region and shipping mode, not company-wide

Bottleneck detection (Section 5) shows delay percentage varies substantially across Order Region and Shipping Mode. Concentrate carrier renegotiation, route audits, and process improvement budget on the specific regions and modes shown to over-index on delay, rather than applying uniform investment across all lanes.

3. Build a full root-cause library across all flagged regions

The East Africa driver analysis (Section 6) demonstrated a repeatable method for isolating the Shipping Mode / Customer Segment / Department / Payment Type / Order Status combinations most associated with delay in a given region. Re-run this analysis for every region flagged in Section 5 to produce a prioritized, region-specific action list for operations leadership.

4. Align staffing and carrier capacity with the delay calendar

Time-based analysis (Section 7) shows delay percentage is not uniform across months, weekdays, or hours. Use the identified peak periods to pre-position warehouse staffing, extend cut-off windows, or secure additional carrier capacity ahead of predictable high-delay windows, rather than reacting after delay percentages spike.

5. Monitor the small high-delay, lower-margin tail

The +3-day delay bucket combines both the lowest mean profit per order (\$20.0) and the smallest order share (3.92%). While not currently material to overall profitability, this segment warrants a standing watch metric to confirm it does not grow as a share of volume.

6. Preserve profitability visibility as an operational guardrail

Since delay length alone does not explain profitability variation (Section 4), any operational change intended to reduce delay should be tracked against the existing profit-per-order baseline to confirm it does not introduce

new cost that erodes the margin the business currently retains even on delayed orders.

Sequencing

Recommendation 1 (deploy the model) and Recommendation 4 (calendar-based staffing) can be implemented immediately using existing analysis. Recommendations 2 and 3 require additional region-by-region analysis using the methodology already validated in this report.

SECTION 10

Conclusion

Across 172,765 analyzed orders, the DataCo supply chain shows a consistent, well-defined pattern rather than an unpredictable operational problem: more than half of all orders ship later than scheduled, delay concentrates in identifiable regions, shipping modes and operational segments, and delay follows a repeatable calendar rhythm. Critically, profitability remains resilient across virtually all delay lengths, which means the business is not facing an urgent margin crisis — it is facing a solvable visibility and prioritization problem.

The predictive model built in Section 8 demonstrates that late delivery can be anticipated with meaningful accuracy using information already captured at order entry, and the bottleneck and root-cause methodologies in Sections 5 and 6 provide a repeatable way to localize the specific operational combinations driving that risk in any region. Combined with the time-based patterns in Section 7, DataCo has, within its own historical data, everything required to move from reactive delay management to a proactive, targeted operational program — without new data collection and without broad, undifferentiated investment.

The recommendations in Section 9 are sequenced so that the highest-confidence, lowest-effort actions (deploying the existing prediction model and aligning staffing to the known delay calendar) can begin immediately, while the region-by-region root-cause expansion builds out over subsequent operating cycles.

Final note

All figures, tables and model results in this report are drawn directly from the source analytical notebook; no supplementary data or external benchmarks were introduced.